

# User Manual for A Simulation Toolbox of Array-Antenna-Received BDS Signals and Composite Interference Datasets

## 1 Software Overview

This tool is a professional intermediate frequency (IF) signal simulation software for Global Navigation Satellite System (GNSS). Developed and packaged as a standalone Windows executable (.exe) based on MATLAB, it requires no MATLAB Runtime installation. Users can configure simulation parameters via an intuitive Graphical User Interface (GUI) and generate complex signal datasets with one click, including satellite signals, spoofing interference, multipath effects, suppression jamming, and multi-antenna array responses.

### 1.1 Core Functions

1. **Satellite Signal Simulation:** Supports generation of multi-satellite signals at the BDS B1I frequency band, with customizable core parameters such as PRN list, carrier-to-noise density ratio ( $C/N_0$ ), sampling rate, and intermediate frequency.
2. **Array Antenna Simulation:** Supports uniform circular array (with/without center element), uniform linear array, planar array, L-shaped array, dual parallel line array, and single antenna; simulates amplitude-phase inconsistency between antenna elements.
3. **Spoofing Interference Simulation:** Supports time-only spoofing and joint time-position spoofing, with optional compliance with the Carrier-Code Consistency Relationship (CCDC).
4. **Multipath Effect Simulation:** Supports adding multiple multipath signals to designated satellites, with uniformly random time delay, Rayleigh fading amplitude, and random perturbation of the direction of arrival (DOA) near the line-of-sight direction.
5. **Suppression Jamming Simulation:** Supports 8 jamming types (single-tone, multi-tone, frequency-swept, pulse, noise FM, noise AM, BPSK, wideband noise), with configurable independent jamming sources.
6. **Data Output:** Supports real/complex signal output; data types include int16, int32, float32, and double. DOA can be recorded and saved as HDF5 files.
7. **Metadata Storage:** Automatically generates HDF5 metadata files containing all simulation parameters, array configurations, multipath/jamming implementation details, and real trajectories for later reproduction and verification.

## 2 Software Installation and Startup

## 2.1 Software Acquisition

This tool is distributed as a single executable file (e.g., GNSS\_Signal\_Generator.exe).

## 2.2 Startup and Shutdown

1. **Startup:** Double-click the .exe file to open the main interface; no installation is required.

2. **Shutdown:** Click the "×" button in the upper-right corner of the window or end the process via Task Manager. If a simulation is running, click the Abort button before closing the window to avoid data file corruption.

## 3 Software Interface Overview

After startup, the main window layout is as follows:

1. **Top Tabs:** Five parameter configuration pages — Basic Parameters, Array Parameters, Spoofing Parameters, Multipath Parameters, Jamming Parameters.

### 2. Bottom Status Bar:

a. **Save Configuration:** Saves all current parameters as a .txt plain-text configuration file.

b. **Run Simulation:** Starts the signal generation task (available only after parameter validation passes).

c. **Abort:** Terminates the ongoing simulation immediately (available only during runtime).

d. **Status Label:** Displays statuses such as "Ready", "Parameter Error", "Simulating", "Completed".

e. **Progress Label:** Displays the number of seconds generated during simulation.

## 4 Detailed Parameter Configuration

### 4.1 Basic Parameters Tab

| Parameter Name              | Description                                                                                 | Value Range/Format        |
|-----------------------------|---------------------------------------------------------------------------------------------|---------------------------|
| Intermediate Frequency (IF) | Center frequency of the simulated signal (after down-conversion by the analog RF front-end) | Real number $\geq 0$ (Hz) |
| Sampling Rate (Fs)          | Signal sampling frequency                                                                   | Real number $> 0$ (Hz)    |

|                                                    |                                                                                                                            |                                       |
|----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| Simulation Duration                                | Total duration of the generated signal                                                                                     | Real number > 0 (s)                   |
| Channel Count                                      | Maximum number of satellites processed in parallel by the receiver (upper limit of actual satellites in the output signal) | Integer $\geq 1$                      |
| Complex Sampling                                   | Check to output I/Q complex signals; uncheck to output real signals                                                        | Boolean                               |
| Carrier-to-Noise Density Ratio (C/N <sub>0</sub> ) | C/N <sub>0</sub> of satellite line-of-sight signals (uniform for all satellites)                                           | Real number (dB-Hz)                   |
| Coordinate System                                  | Adopted coordinate system (fixed to BDCS)                                                                                  | BDCS                                  |
| Time System                                        | Adopted time system (fixed to BDST)                                                                                        | BDST                                  |
| PRN List                                           | List of satellite PRN numbers to simulate                                                                                  | e.g., [6,7,9,10] (PRN 1–63)           |
| Enable Satellite Signals                           | Whether to generate satellite line-of-sight signals (if disabled, only noise/jamming/multipath are generated)              | Boolean                               |
| Data Format                                        | Data type of the output binary file, affecting dynamic range and file size                                                 | int16 / int32 / float32 / double      |
| File Save Path                                     | Storage directory of output files (write permission required)                                                              | Valid folder path                     |
| File Name                                          | Main file name of the output (without extension)                                                                           | Valid file name (excluding `/:*?"<>`) |
| Record DOA Information                             | Whether to save DOA information as a separate HDF5 file                                                                    | Boolean                               |

#### Detailed Parameter Descriptions:

- Complex Sampling:** If checked, the output file is stored in interleaved format I<sub>1</sub>, Q<sub>1</sub>, I<sub>2</sub>, Q<sub>2</sub>, ...; if unchecked, only the real part is output.
- PRN List:** Input must be a MATLAB-style array, e.g., [1,5,8,12]. Satellites with PRNs invisible at the current time (elevation < 5°) will not be simulated.

3. **Sampling Rate (Fs):** Must satisfy the Nyquist sampling theorem. For BDS B1I signals, the sampling rate shall be no less than 4.092 MHz.

## 4.2 Array Parameters Tab

| Parameter Name                             | Description                                                                              | Value/Option         |
|--------------------------------------------|------------------------------------------------------------------------------------------|----------------------|
| Array Type                                 | Geometric arrangement of the antenna array                                               | See table below      |
| Element Count                              | Total number of antenna elements                                                         | Integer $\geq 1$     |
| Array Radius                               | Array radius (in wavelengths corresponding to the signal center frequency)               | Real number $\geq 0$ |
| Channel Inconsistency                      | Whether to simulate amplitude-phase inconsistency of receiving channels for each element | Boolean              |
| Amplitude Inconsistency Standard Deviation | Standard deviation of amplitude differences between elements (dB)                        | Real number $\geq 0$ |
| Phase Inconsistency Standard Deviation     | Standard deviation of phase differences between elements (deg)                           | 0 ~ 180              |

### Detailed Array Type Descriptions:

| Option                               | Array Type                                      | Element Arrangement Description                                                     |
|--------------------------------------|-------------------------------------------------|-------------------------------------------------------------------------------------|
| Uniform Circular Array (with center) | Uniform Circular Array (with center element)    | 1 center element + remaining elements uniformly distributed on a circle of radius R |
| Uniform Circular Array (no center)   | Uniform Circular Array (without center element) | All elements uniformly distributed on a circle of radius R                          |
| Uniform Linear Array                 | Uniform Linear Array                            | Equally spaced along the positive X-axis (spacing = radius $\times$ wavelength)     |
| Planar Array                         | Planar Array                                    | Uniform rectangular planar array, with rows and columns calculated                  |

|                          |                          |                                                                                   |
|--------------------------|--------------------------|-----------------------------------------------------------------------------------|
|                          |                          | automatically                                                                     |
| L-shaped Array           | L-shaped Array           | Arranged in an L-shape along positive X and Y axes (origin as the common element) |
| Dual Parallel Line Array | Dual Parallel Line Array | Two parallel linear arrays with spacing equal to the radius                       |
| Single Antenna           | Single Antenna           | Single antenna (element count forced to 1; other array parameters invalid)        |

### 4.3 Spoofing Parameters Tab

| Parameter Name             | Description                                                                               | Value/Format                                                          |
|----------------------------|-------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| Spoofing Mode              | Spoofing type selection                                                                   | 0-No spoofing / 1-Time-only spoofing / 2-Joint time-position spoofing |
| Comply with CCDC           | Whether the spoofing signal carrier and code phase maintain a fixed multiple relationship | Check=Comply (non-frequency-locked)                                   |
| Maximum SSR                | Maximum power ratio of spoofing signal to real signal                                     | Real number (dB)                                                      |
| Time Spoofing Interval     | Start and end time of time-only spoofing (or time phase of joint spoofing)                | [Start, End] (s)                                                      |
| Position Spoofing Interval | Start and end time of position spoofing pull-in (valid only for Mode 2)                   | [Start, End] (s)                                                      |
| Position Offset            | Final position offset of spoofing (East-North-Up coordinate system)                       | [East, North, Up] (m)                                                 |
| Spoofing Position          | Position of the spoofing interference source in ECEF coordinate system                    | [x, y, z] (m)                                                         |

#### Detailed Spoofing Mode Behavior:

1. **Mode 0:** No spoofing signal.
2. **Mode 1** (Time-only Spoofing): Within the time spoofing interval, the code phase of

the spoofing signal changes in a trapezoidal rate curve, with a total pull-in of  $\pm 1.5$  chips (sign determined by the east component of the position offset). The spoofing amplitude fades in linearly in the first 10 seconds of the interval, fades out linearly in the last 10 seconds, and remains at maximum SSR within the interval.

3. **Mode 2** (Joint Time-Position Spoofing): Executes the time spoofing phase first, then enters the position spoofing phase, with real-time calculation of the code phase offset of the spoofing signal.

#### CCDC Relationship Description:

1. **Comply with CCDC** (checked): Spoofing signal carrier frequency pull-in rate = code frequency pull-in rate  $\times \beta$  ( $\sim 763$  for B1I band), making the spoofing signal harder to detect.

2. **Non-compliant with CCDC** (unchecked): Carrier frequency remains constant or outputs a fixed offset, belonging to frequency-locked spoofing.

### 4.4 Multipath Parameters Tab

| Parameter Name                  | Description                                                                                    | Value/Format       |
|---------------------------------|------------------------------------------------------------------------------------------------|--------------------|
| Enable Multipath                | Whether to add multipath signals                                                               | Boolean            |
| Multipath Satellite PRN List    | Designate satellites for multipath addition (empty [] = all visible satellites)                | [prn1, prn2, ...]  |
| Multipath-to-Signal Ratio (MSR) | Ratio of total power of all multipath signals to line-of-sight signal power                    | Real number (dB)   |
| Multipath Path Count            | Number of multipath signals generated per satellite                                            | Integer $\geq 1$   |
| Time Delay Range                | Time delay range of multipath signals relative to line-of-sight signals (uniform distribution) | [Min, Max] (chips) |
| Multipath Activation Interval   | Start and end time of multipath signal existence                                               | [Start, End] (s)   |

#### Multipath Signal Generation Mechanism:

1. Time delay of each multipath follows a uniform distribution within the specified range.

2. Amplitude of each multipath follows a Rayleigh distribution, with the expected total power equal to the set MSR.

3. Phase of each multipath is uniformly distributed in  $[0, 2\pi]$ , superimposed with carrier phase offset caused by time delay.

4. DOA of each multipath is randomly perturbed near the line-of-sight direction: azimuth  $\pm 30^\circ$ , elevation  $\pm 15^\circ$ .

5. Multipath parameters are randomly initialized at the start of the interval and remain unchanged during the interval.

#### 4.5 Jamming Parameters Tab

| Parameter Name          | Description                                                                         | Value/Format             |
|-------------------------|-------------------------------------------------------------------------------------|--------------------------|
| Enable Jamming          | Whether to add suppression jamming                                                  | Boolean                  |
| Jamming Source Count    | Number of independent jamming sources                                               | Integer $\geq 1$         |
| Jamming Type            | Type of each jamming source, separated by English commas                            | See supported type list  |
| JSR Array               | Jamming-to-signal ratio (JSR) of each jamming source (matches jamming source count) | [JSR1, JSR2, ...] (dB)   |
| Start Time Array        | Activation start time of each jamming source                                        | [t1, t2, ...] (s)        |
| End Time Array          | Deactivation time of each jamming source                                            | [t1, t2, ...] (s)        |
| Jamming Source Position | Position of each jamming source in ECEF coordinate system                           | [x1,y1,z1; x2,y2,z2] (m) |

Supported Jamming Types (case-sensitive):

tone (single-tone jamming), multi\_tone (multi-tone jamming), sweep (frequency-swept jamming), fm\_noise (noise FM jamming), pulse (pulse jamming), bpsk (BPSK jamming), am\_noise (noise AM jamming), wideband\_noise (wideband noise jamming)

Jamming Signal Parameters (automatically enabled by jamming type):

| Parameter Name   | Associated Jamming Type | Description                   | Typical Value |
|------------------|-------------------------|-------------------------------|---------------|
| Frequency Offset | All types               | Carrier offset relative to IF | 500e3 (Hz)    |

|                           |                                 |                                              |                          |
|---------------------------|---------------------------------|----------------------------------------------|--------------------------|
| Bandwidth                 | sweep, fm_noise, wideband_noise | Sweep/noise bandwidth                        | 4.092e6 (Hz)             |
| Pulse Width               | pulse                           | Duration of a single pulse                   | 1e-3 (s)                 |
| Pulse Period              | pulse                           | Pulse repetition period (must > pulse width) | 5e-3 (s)                 |
| Multi-tone Frequency List | multi_tone                      | Offset list of each tone                     | [-100e3,50e3,150e3] (Hz) |
| Sweep Period              | sweep                           | Linear sweep repetition period               | 5e-3 (s)                 |
| BPSK Code Rate            | bpsk                            | Pseudo-random code rate                      | 2.046e6 (Hz)             |
| Noise AM Bandwidth        | am_noise                        | Modulation noise bandwidth                   | 4.096e6 (Hz)             |
| FM Modulation Bandwidth   | fm_noise                        | Noise FM modulation bandwidth                | 1.023e6 (Hz)             |

**Configuration Example:** 2 jamming sources: single-tone (JSR 60dB) + pulse (JSR 55dB)

- Jamming Source Count: 2
- Jamming Type: tone,pulse
- JSR Array: [60, 55]
- Start/End Time: [30, 40]/[60, 70]
- Jamming Source Position: [x1,y1,z1; x2,y2,z2]

## 5 Operation Procedure Guide

### 5.1 First-Time Use Steps

1. Launch the software and check default parameters;
2. Modify basic parameters (sampling rate, duration, save path, etc.);
3. Configure array parameters (skip for single antenna);
4. Enable spoofing, multipath, jamming as needed and configure parameters;
5. Click Save Configuration to generate a .txt configuration file for reuse.



## 5.2 Parameter Validation and Error Prompts

1. The software validates parameter legitimacy in real time; invalid items are marked red with hover prompts;
2. The status bar displays a brief error summary, and the Run Simulation button is disabled;
3. The status returns to "Ready" and the button is enabled after errors are corrected.

## 5.3 Running Simulation

1. Confirm the status bar shows "Ready";
2. Click Run Simulation; the status bar switches to "Simulating";
3. Click Abort to terminate early (stops after the current second is generated).

## 5.4 Generated File Description

| File Type        | File Naming Rule                | Description                                    |
|------------------|---------------------------------|------------------------------------------------|
| Signal Data File | FileName_elemX.bin/FileName.bin | Binary signal sampling data                    |
| DOA Record File  | FileName_DoA_records.h5         | DOA information (generated if checked)         |
| Metadata File    | FileName_metadata.h5            | Full simulation parameters and trajectory data |

## 6 Output File Format Details

### 6.1 Signal Data File (.bin)

1. Real signal: Stored in sampling point order in the selected format;
2. Complex signal: Stored in interleaved format  $I_1, Q_1, I_2, Q_2, \dots$

### 6.2 DOA Record File (.h5)

| Dataset Path | Data Type | Description                                           |
|--------------|-----------|-------------------------------------------------------|
| /time_s      | double    | Recording time of signal DOA information (seconds)    |
| /prn         | double    | Satellite PRN number corresponding to DOA information |

|                |        |                                                               |
|----------------|--------|---------------------------------------------------------------|
| /index         | double | Path/jamming source number                                    |
| /azimuth_deg   | double | Azimuth (deg)                                                 |
| /elevation_deg | double | Elevation (deg)                                               |
| /type_code     | int8   | Signal type<br>(1=Satellite/2=Spoofing/3=Multipath/4=Jamming) |
| /steering_real | double | Real part of steering vector                                  |
| /steering_imag | double | Imaginary part of steering vector                             |

## 6.3 Metadata File (.h5)

```

/ (Root)
├── info/          # Tool and generation information
├── parameters/    # Full simulation parameters
│   ├── basic/     # Basic parameters
│   ├── array/     # Array parameters
│   ├── array_mismatch/ # Channel inconsistency parameters
│   ├── spoof/     # Spoofing parameters and trajectories
│   ├── multipath/  # Multipath parameters and implementation details
│   ├── jamming/   # Jamming parameters
│   └── trajectory/ # Real carrier trajectory

```

The software internally uses the default circular trajectory file circle.csv.

## 7 FAQs and Precautions

### 7.1 Common Parameter Errors and Solutions

| Error Symptom             | Cause                            | Solution                                          |
|---------------------------|----------------------------------|---------------------------------------------------|
| PRN list marked red       | Invalid format/out of 1–63 range | Enter valid PRNs in [1,2,3] format                |
| Invalid spoofing interval | Abnormal start/end time          | Ensure Start < End and within simulation duration |
| JSR length mismatch       | Inconsistent with jamming        | Keep the two quantities                           |

|                        | source count                                | consistent                                   |
|------------------------|---------------------------------------------|----------------------------------------------|
| Invalid path/file name | No permission/containing special characters | Select a valid path and use legal characters |

## 7.2 Data Interpretation Notes

1. Signal files are pure binary streams with no header information and must be parsed according to simulation parameters;
2. DOA records are sampled at 1-second intervals, not corresponding to each sampling point;
3. The actual simulated PRNs may differ from the preset list due to elevation filtering.